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(45) **Date of Patent:** Sep. 9, 2025

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- Primary Examiner — Eva Y Montalvo

- Assistant Examiner — Jiyoungh Oh

- (74) *Attorney, Agent, or Firm* — MUNCY, GEISSLER,
OLDS & LOWE, P.C.

- (57) **ABSTRACT**

- The present application provides a method for manufacturing a memory device having a word line (WL) with dual conductive materials. The method includes steps of providing a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate; forming a recess extending from the surface into the semiconductor substrate; disposing a first insulating layer conformal to the recess; disposing a first conductive material within the recess and surrounded by the first insulating layer; removing a portion of the first conductive material to form a first conductive member; disposing a second insulating layer within the recess and conformal to the first insulating layer and the first conductive member; and disposing a second conductive material within the recess and surrounded by the second insulating layer to form a second conductive member adjacent to the first conductive member.

- 16 Claims, 20 Drawing Sheets**

- [illegible]

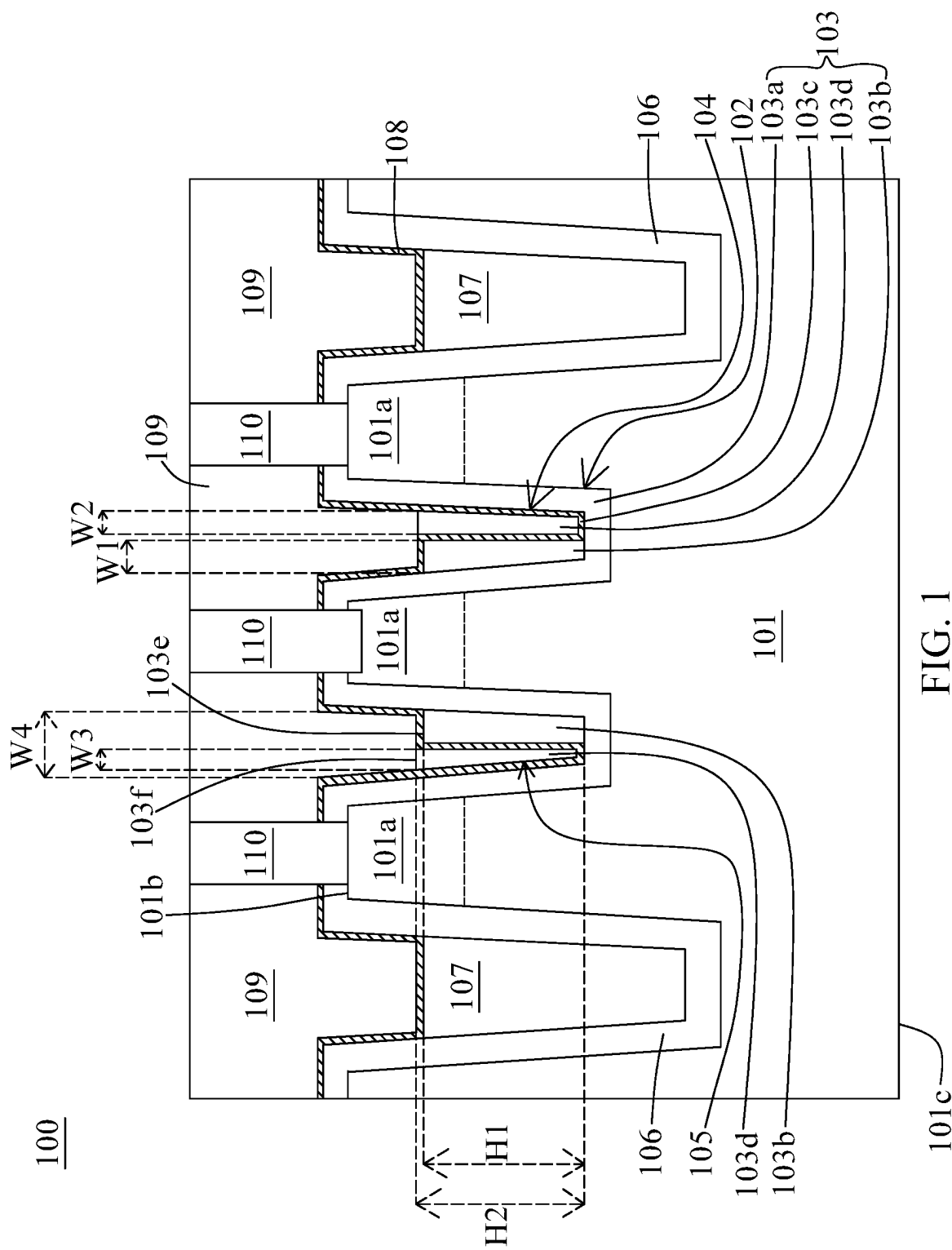
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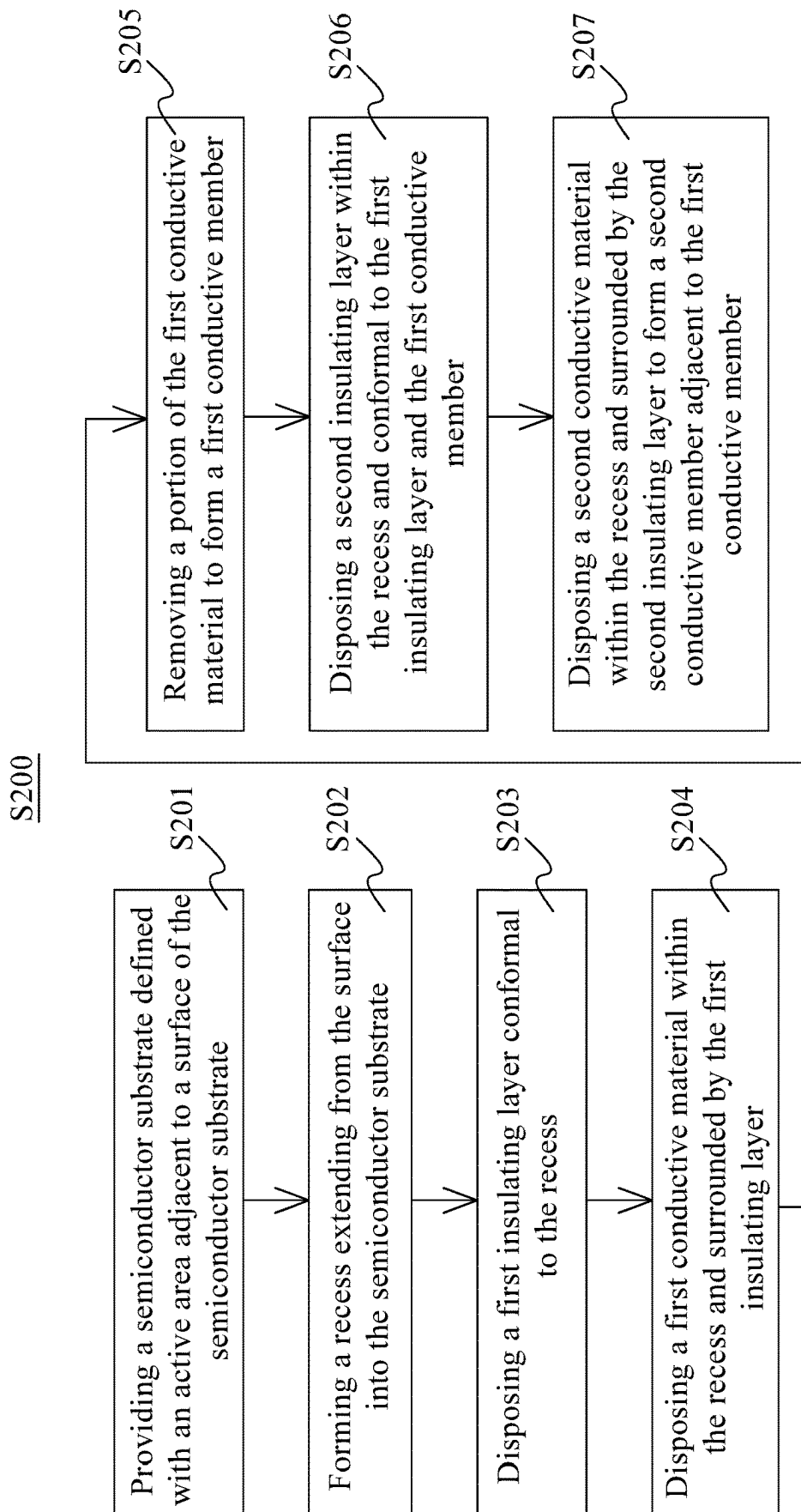


FIG. 2

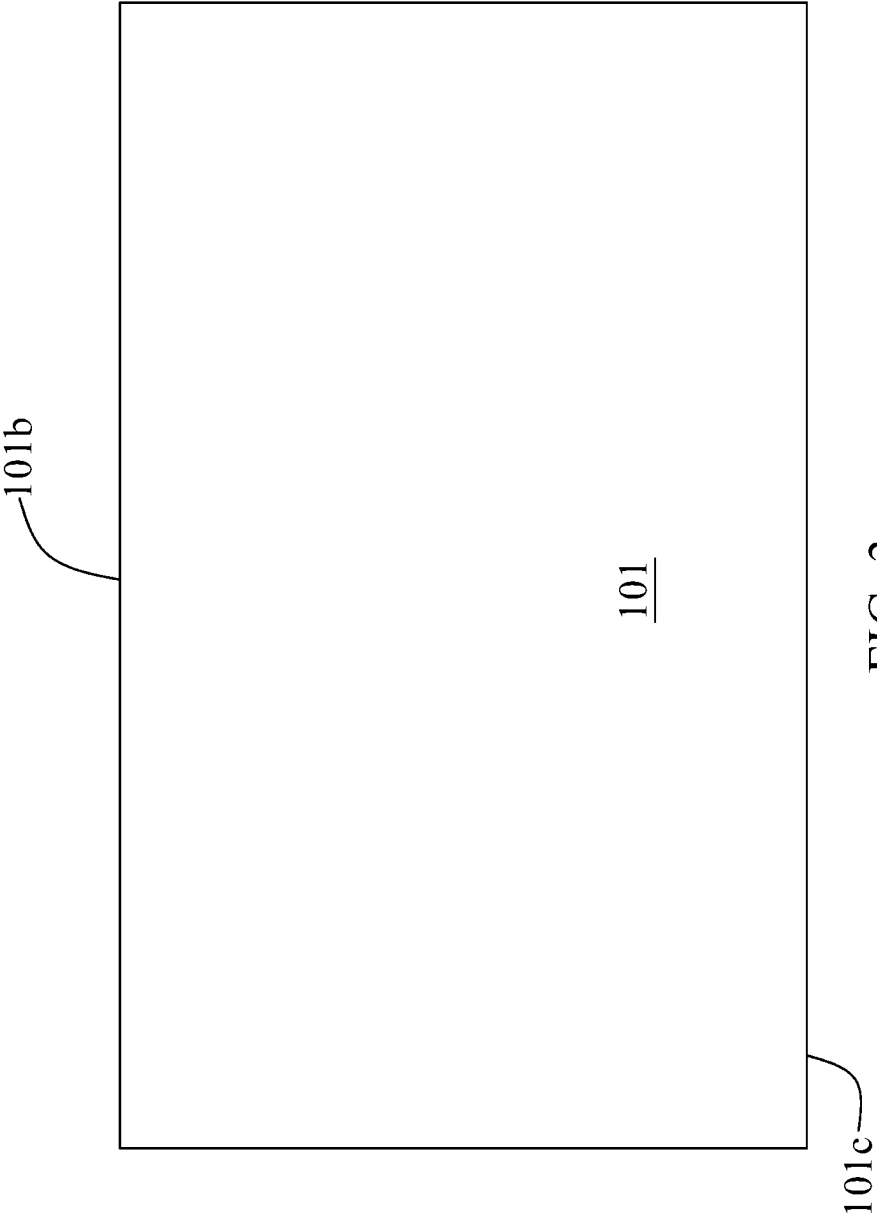
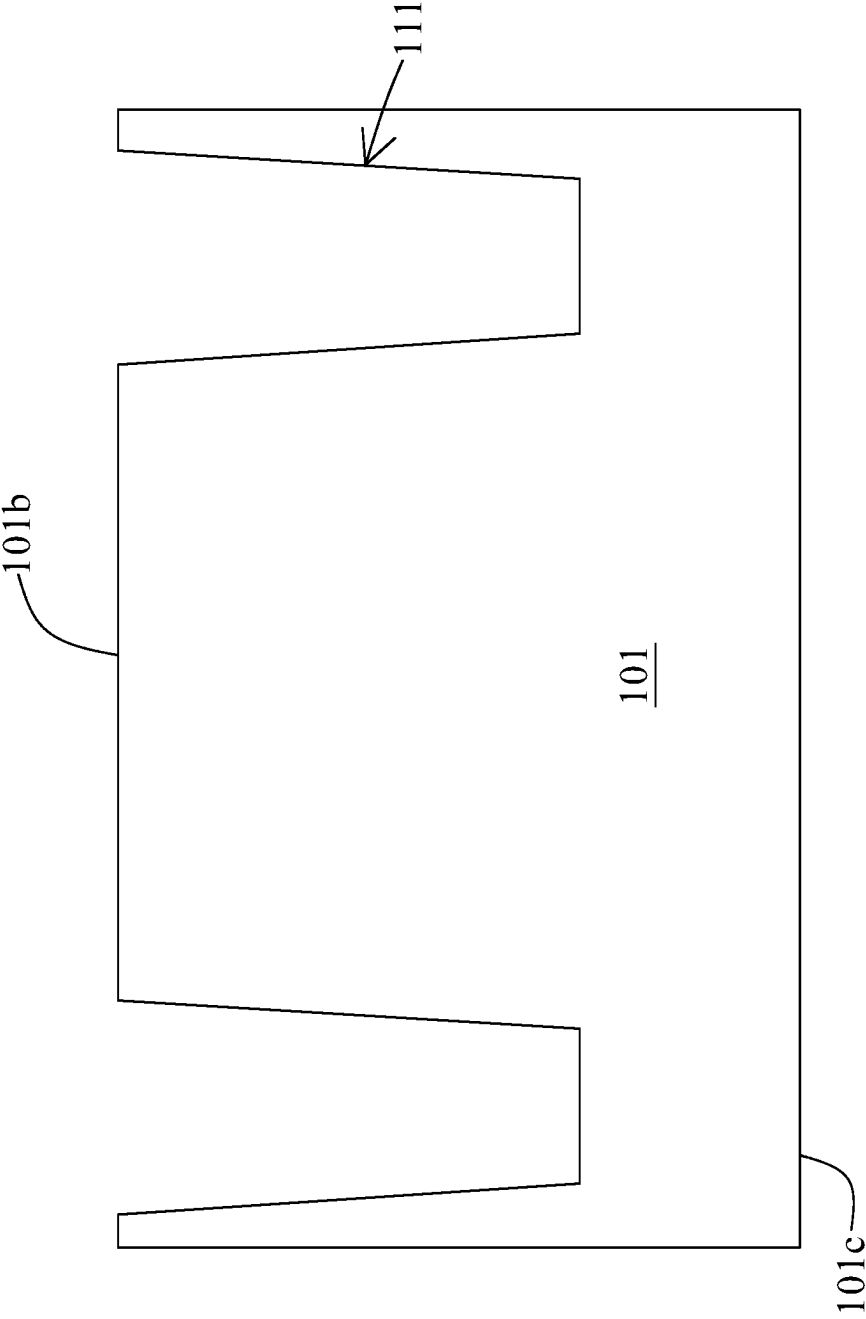
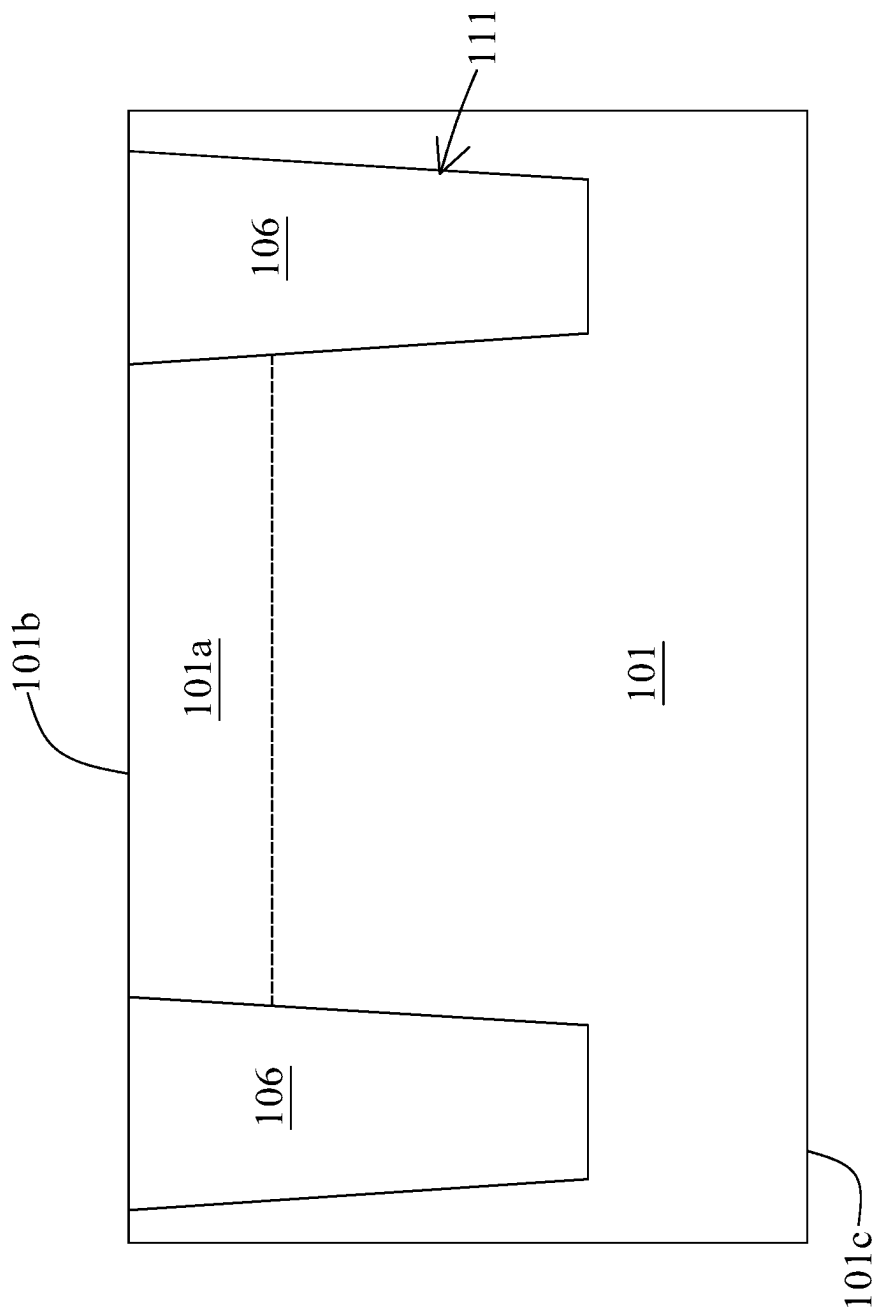
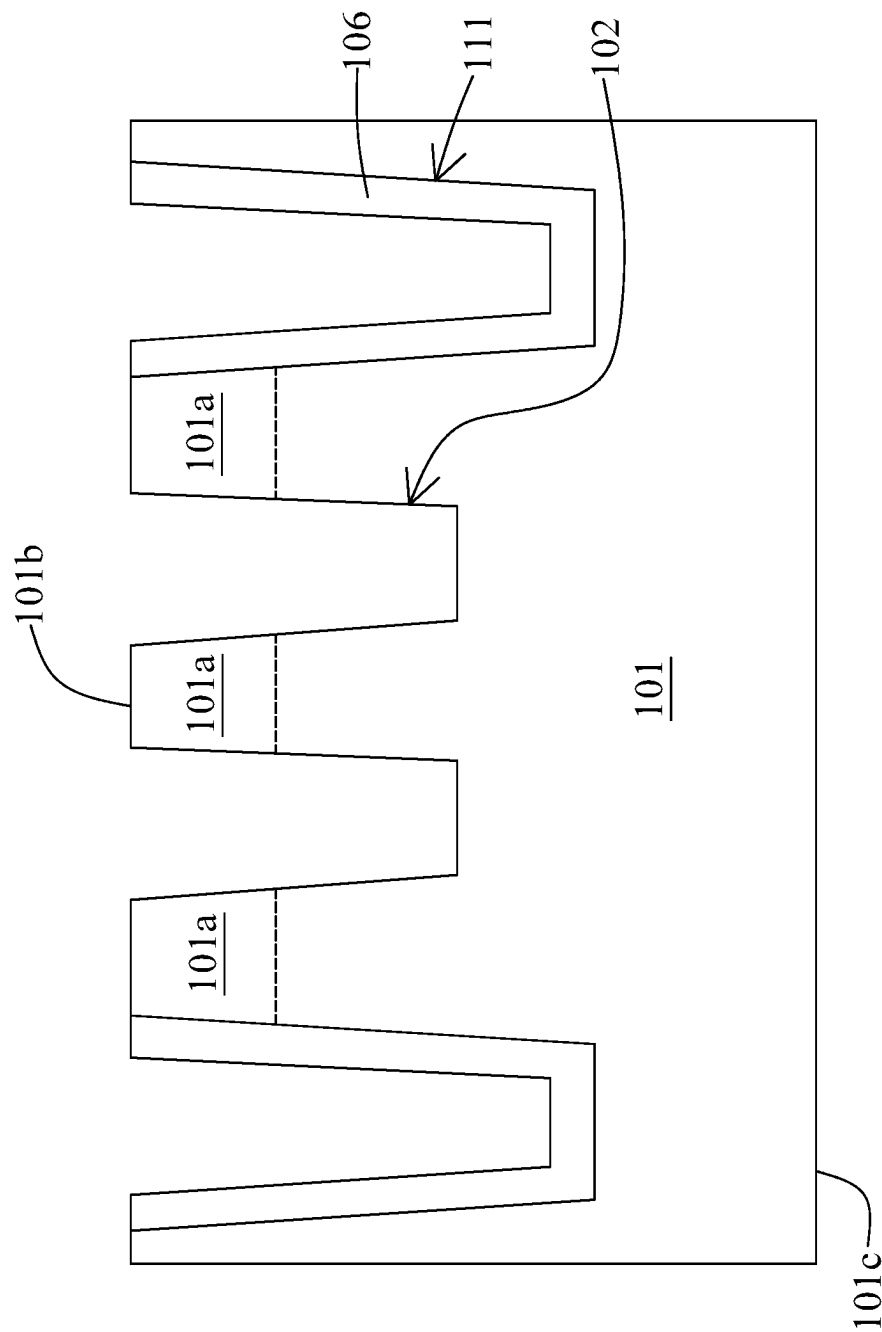
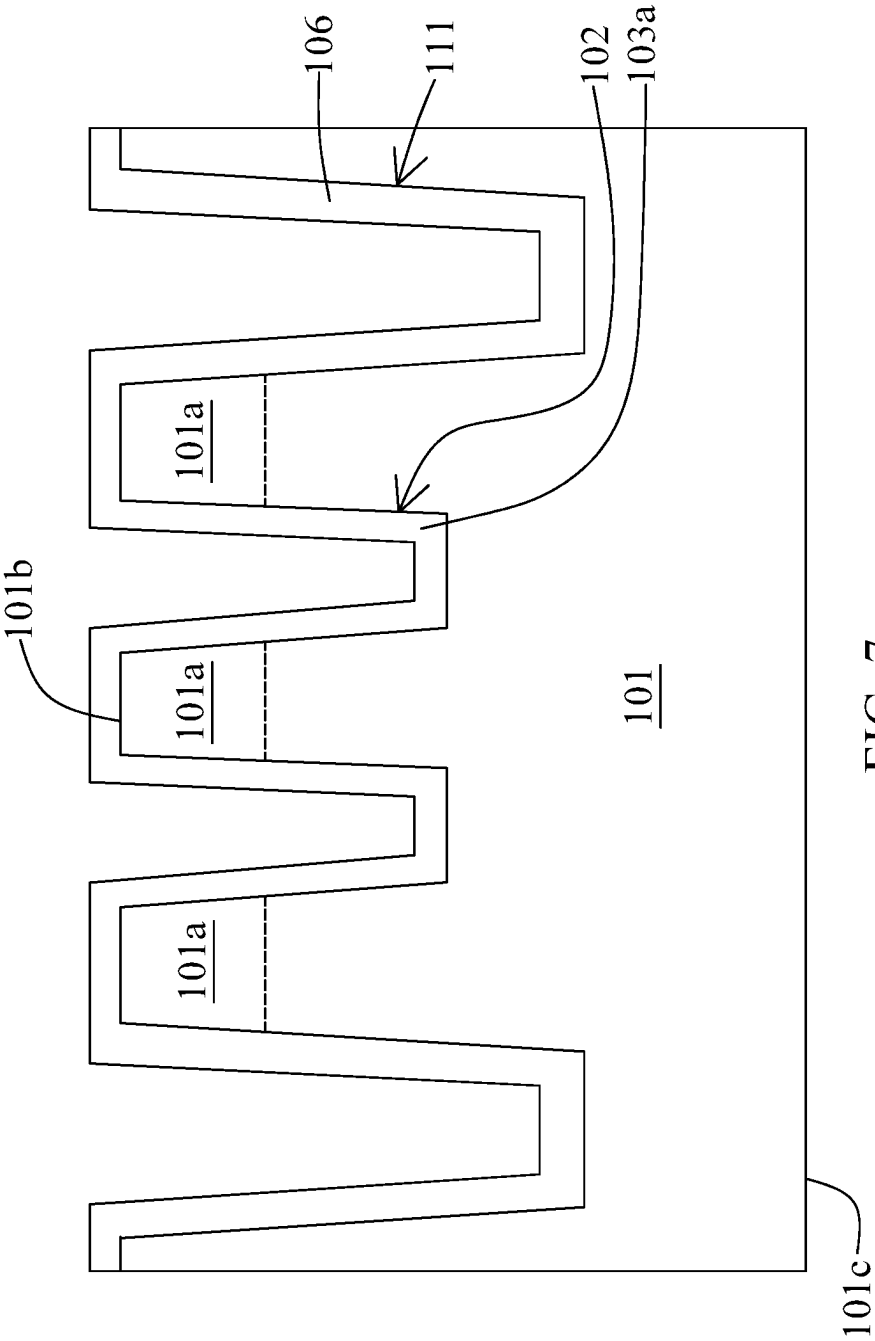


FIG. 3









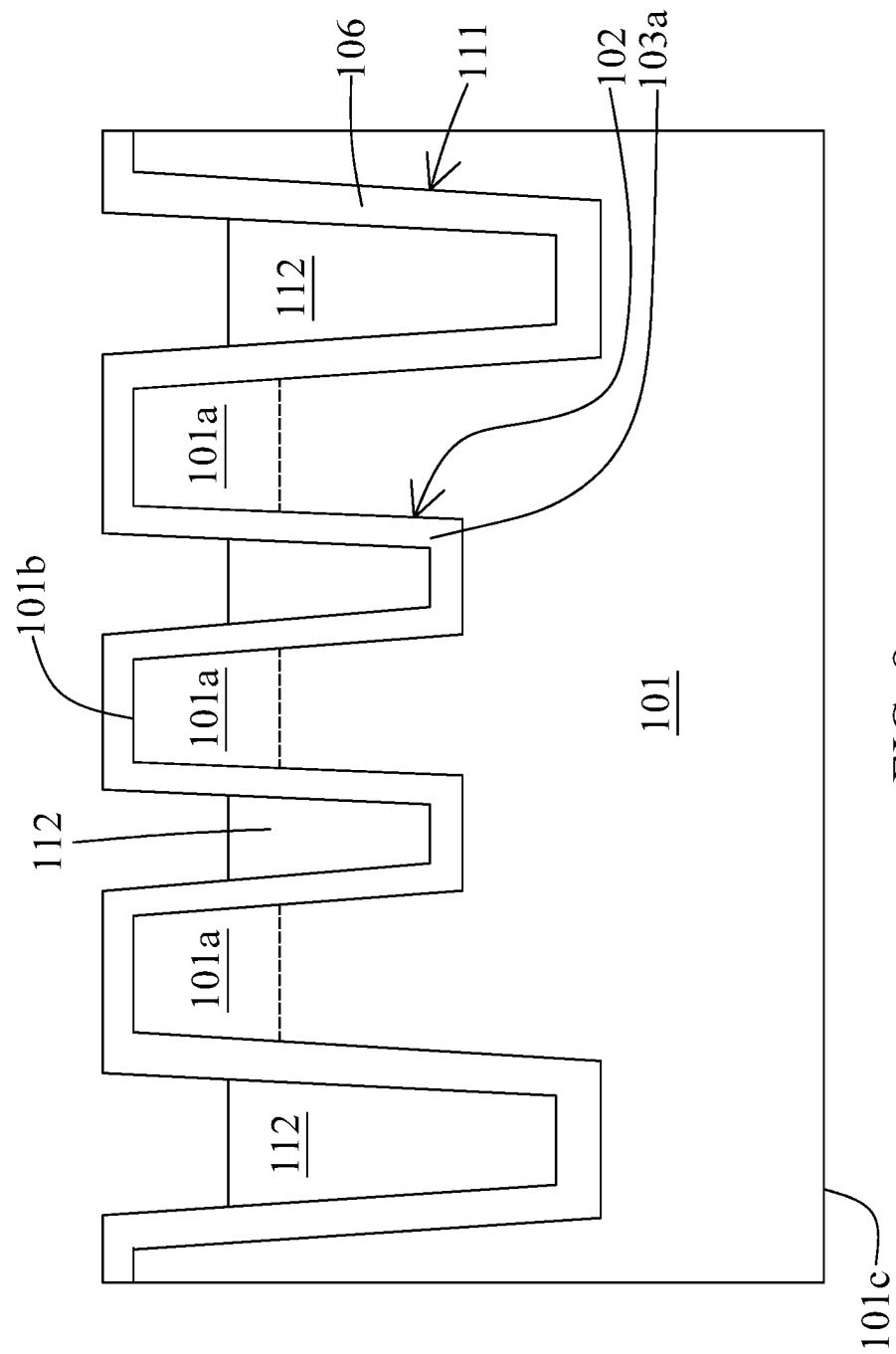


FIG. 8

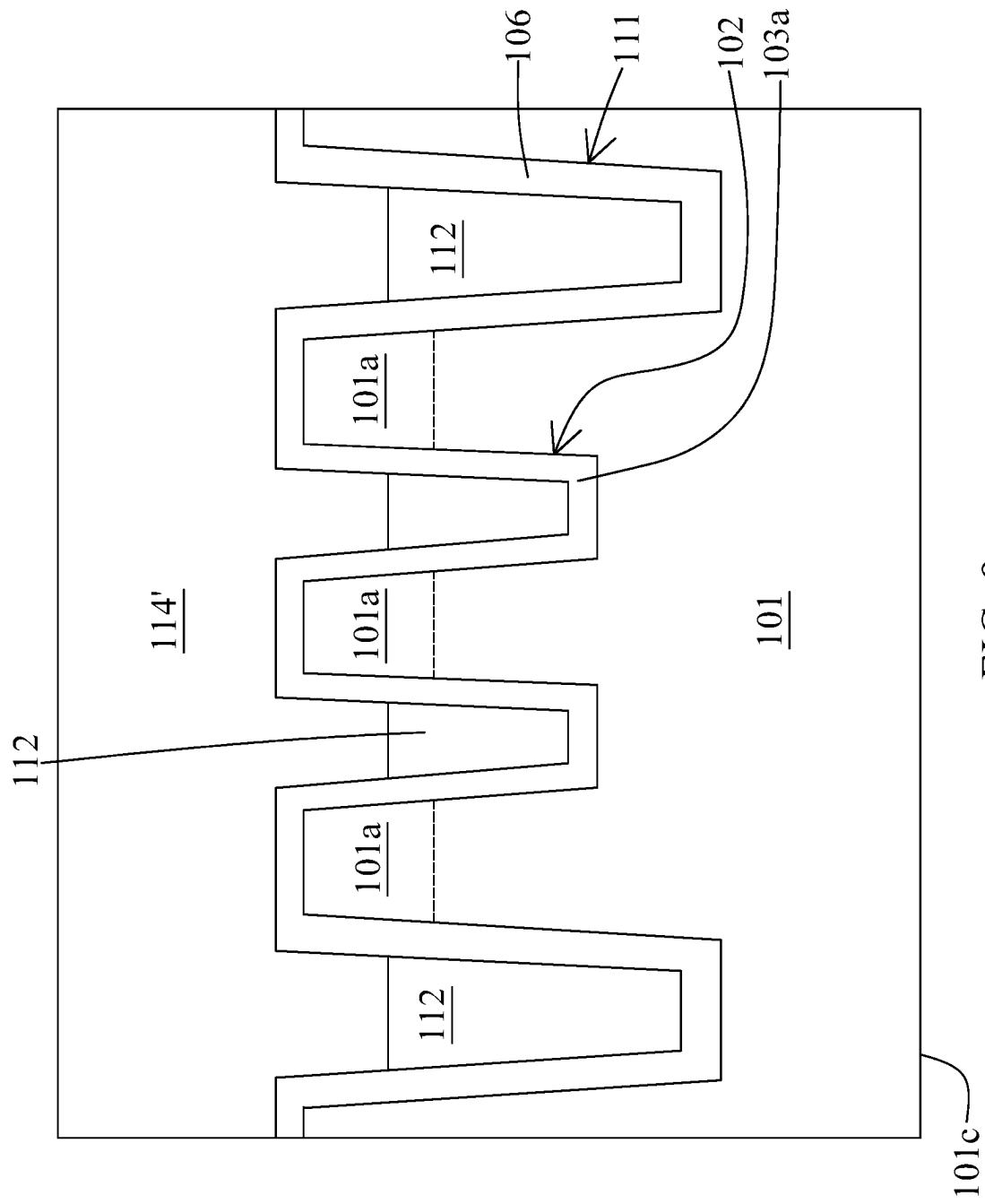


FIG. 9

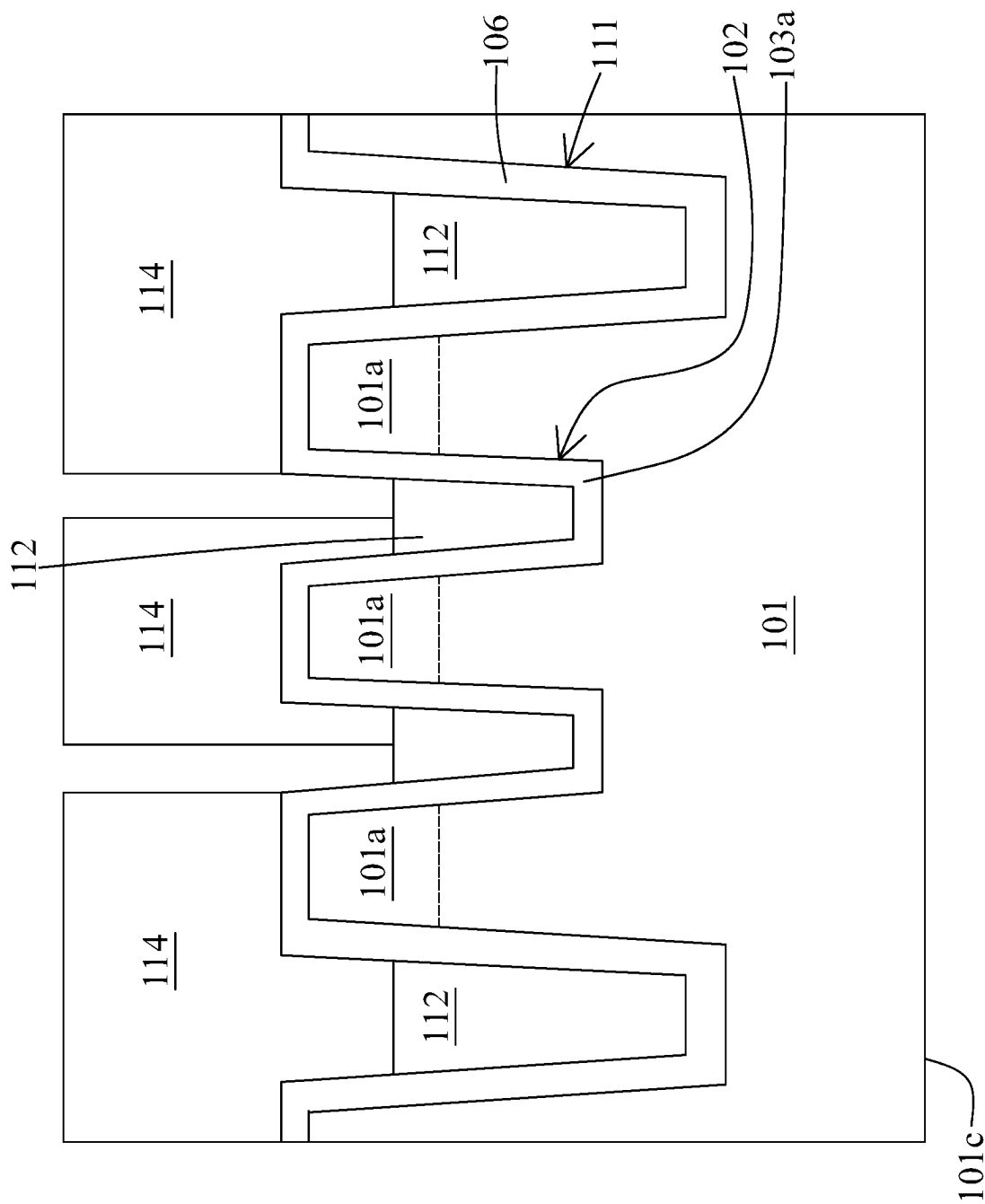


FIG. 10

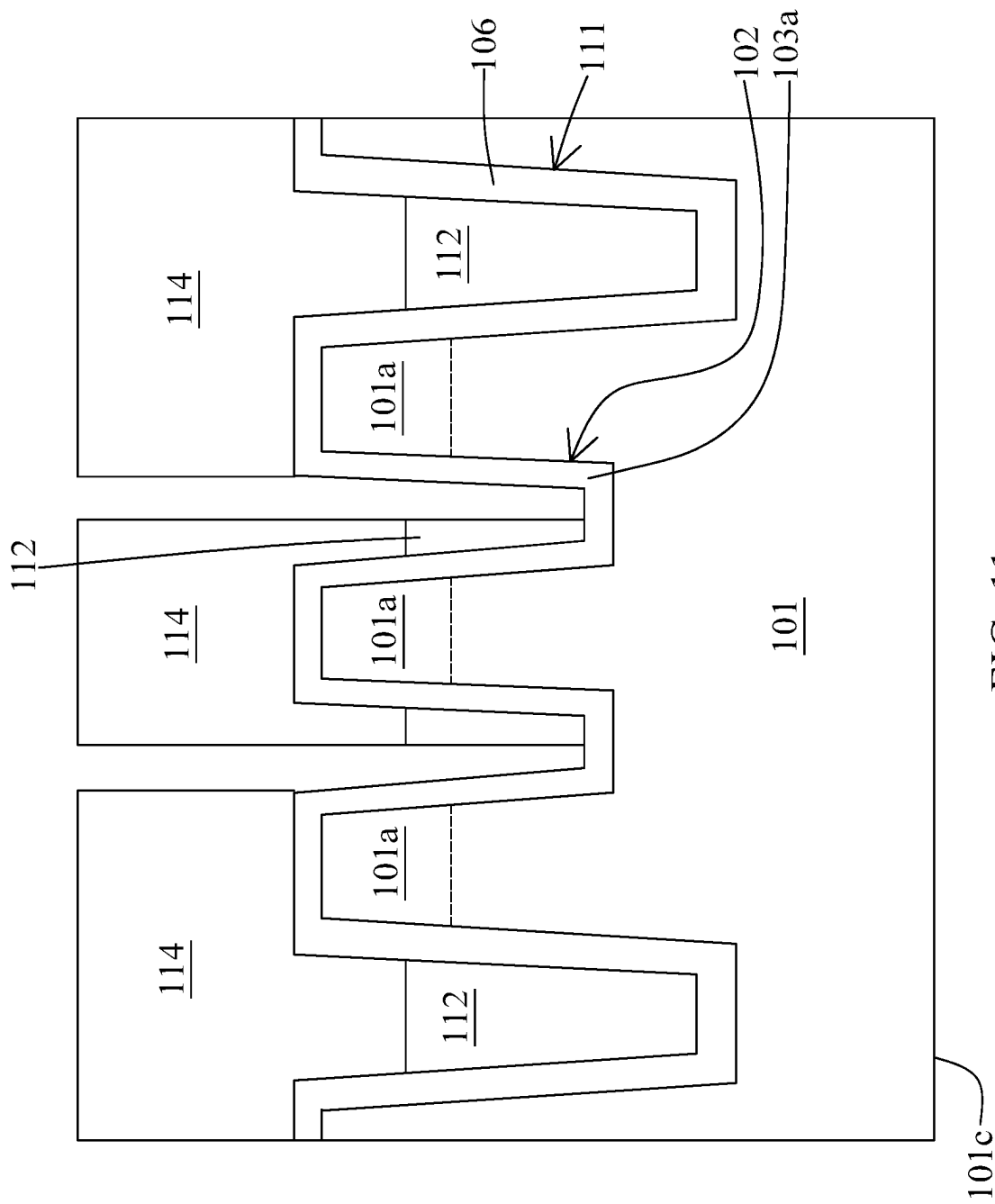


FIG. 11

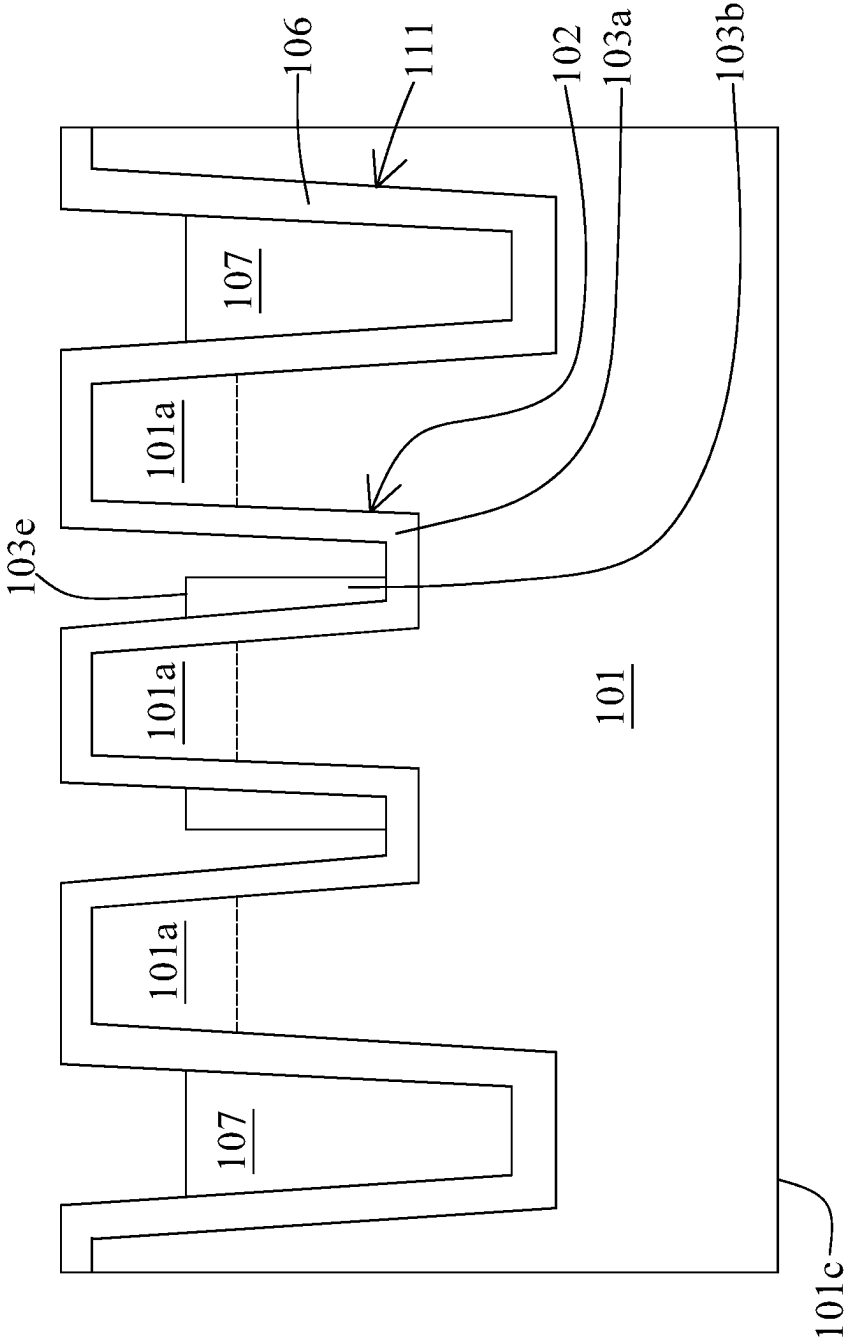


FIG. 12

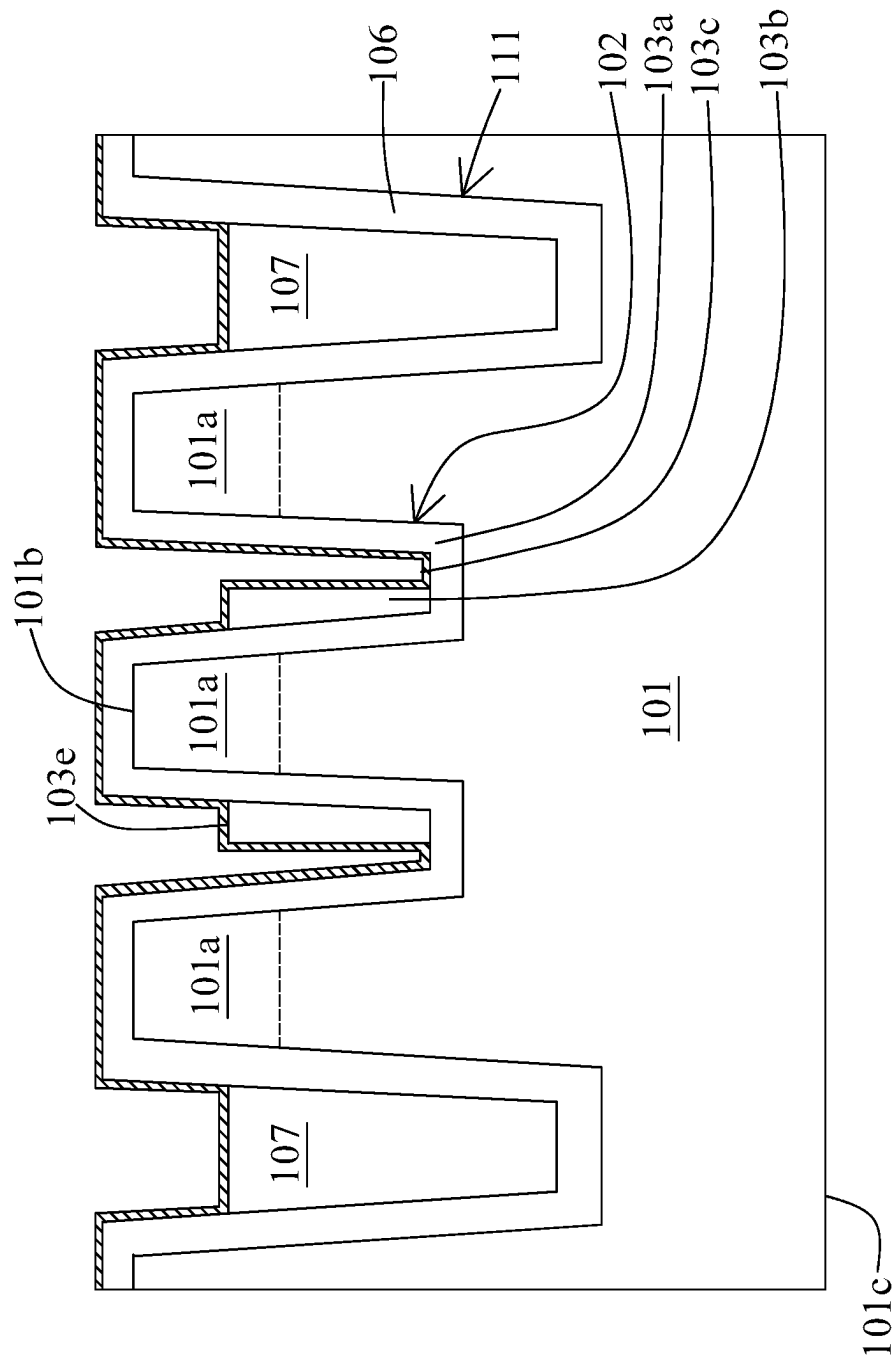
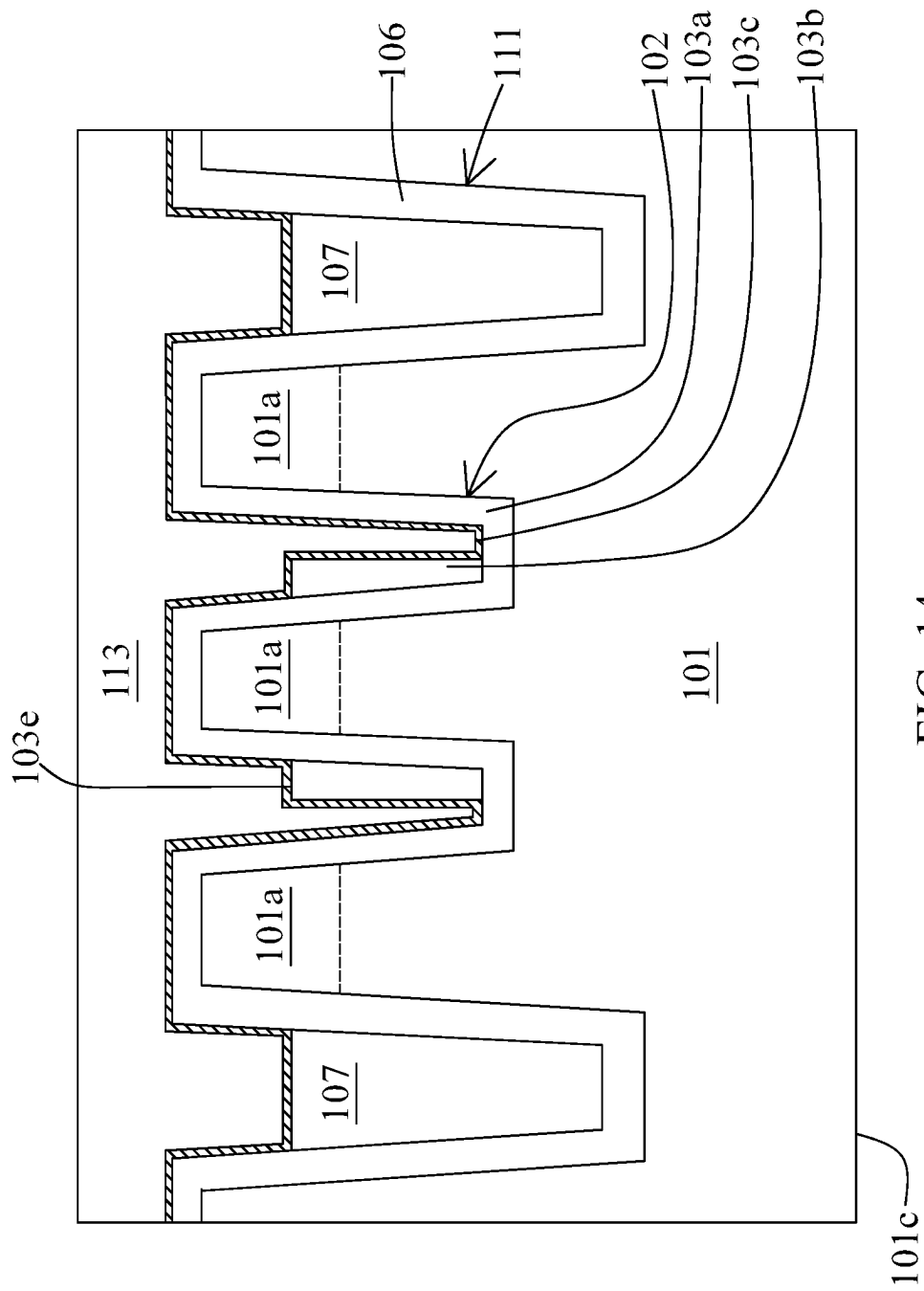


FIG. 13



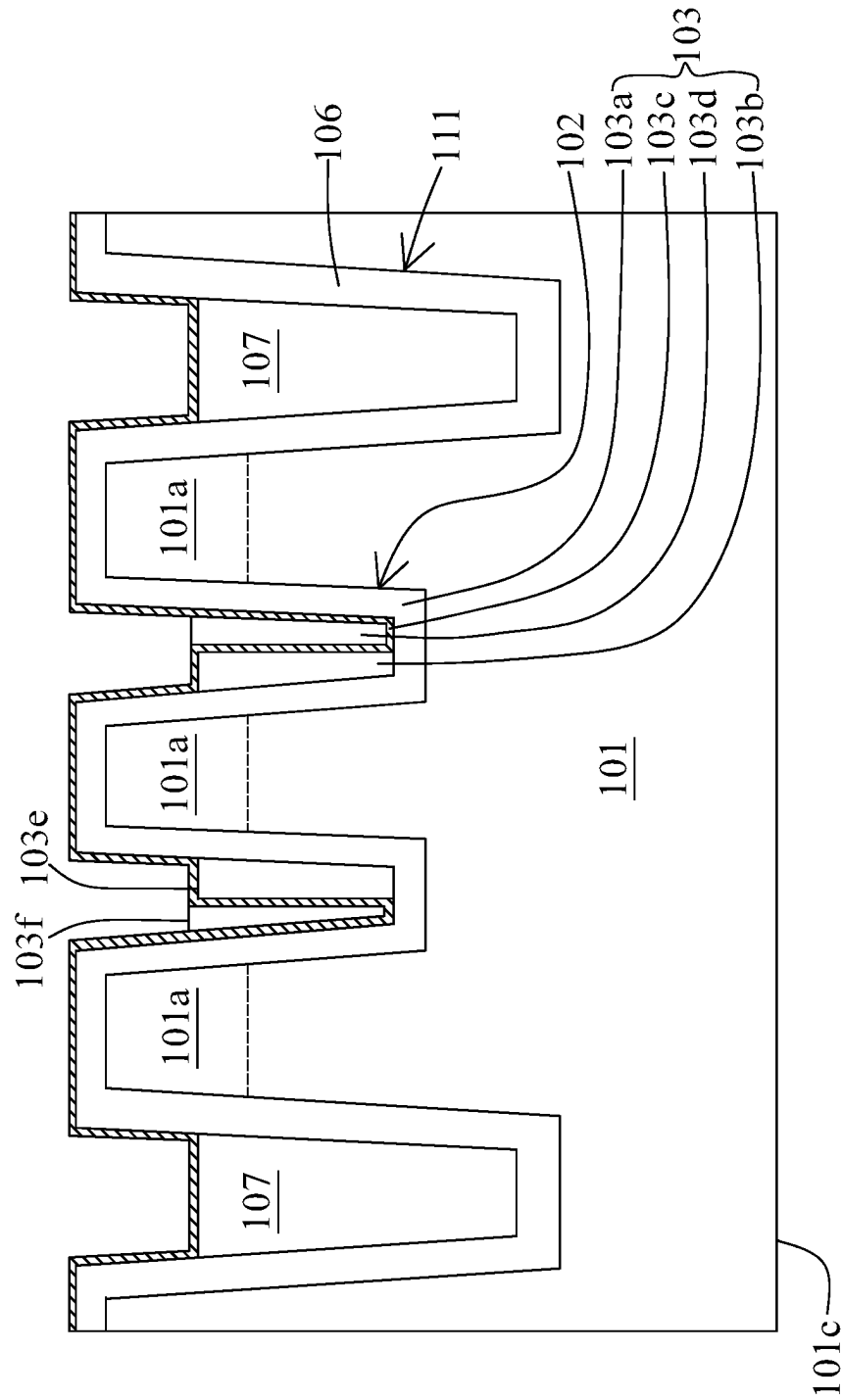


FIG. 15

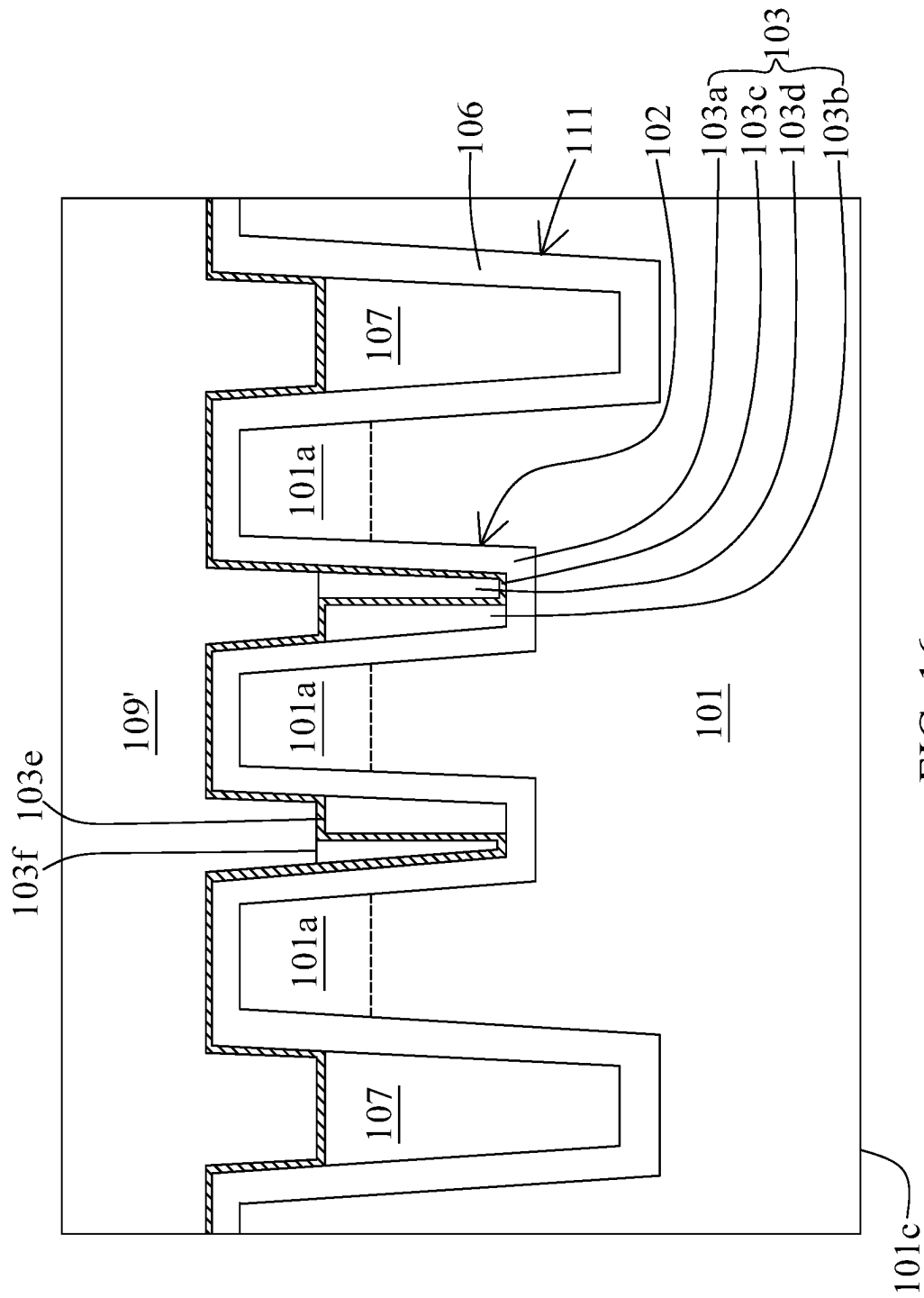


FIG. 16

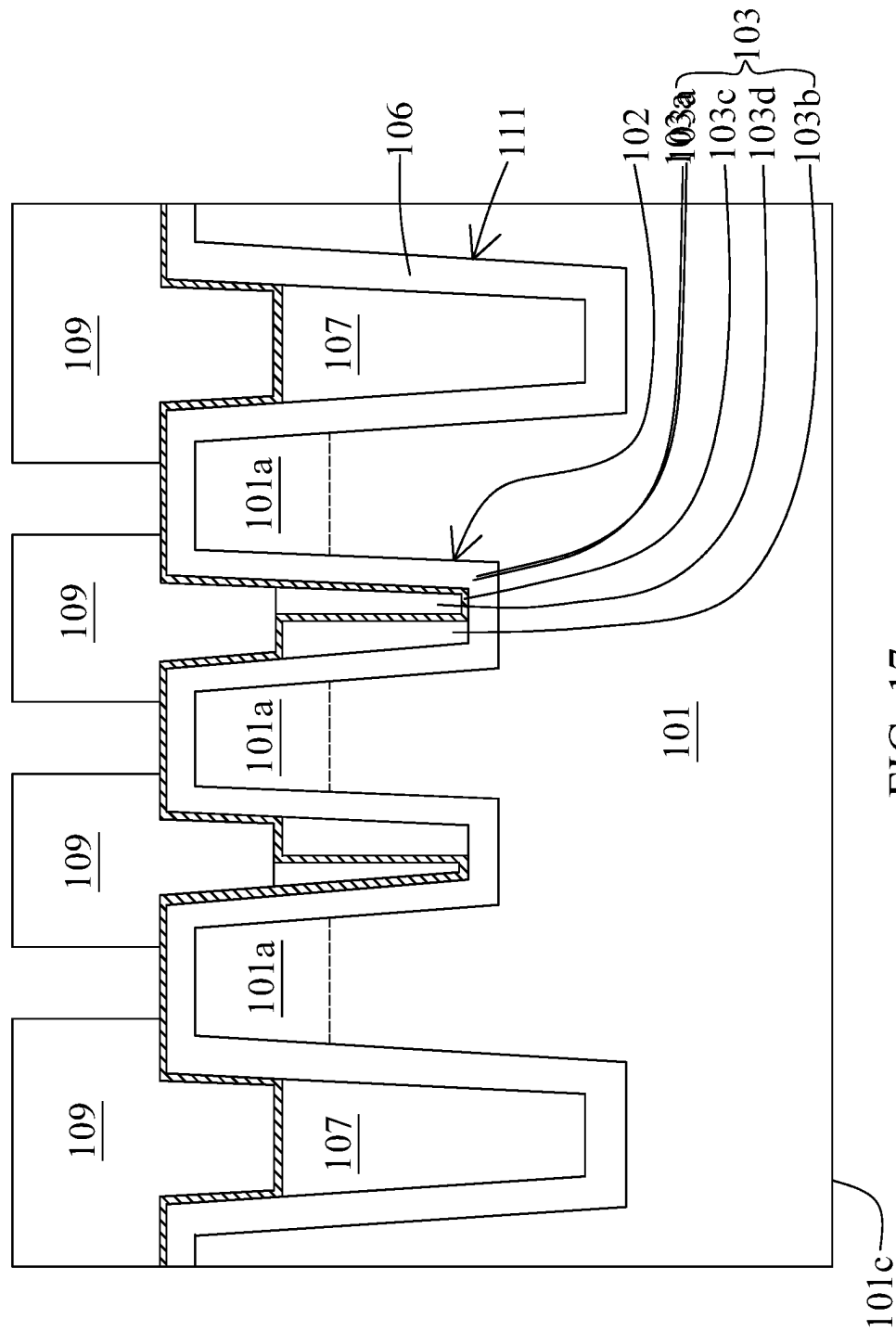


FIG. 17

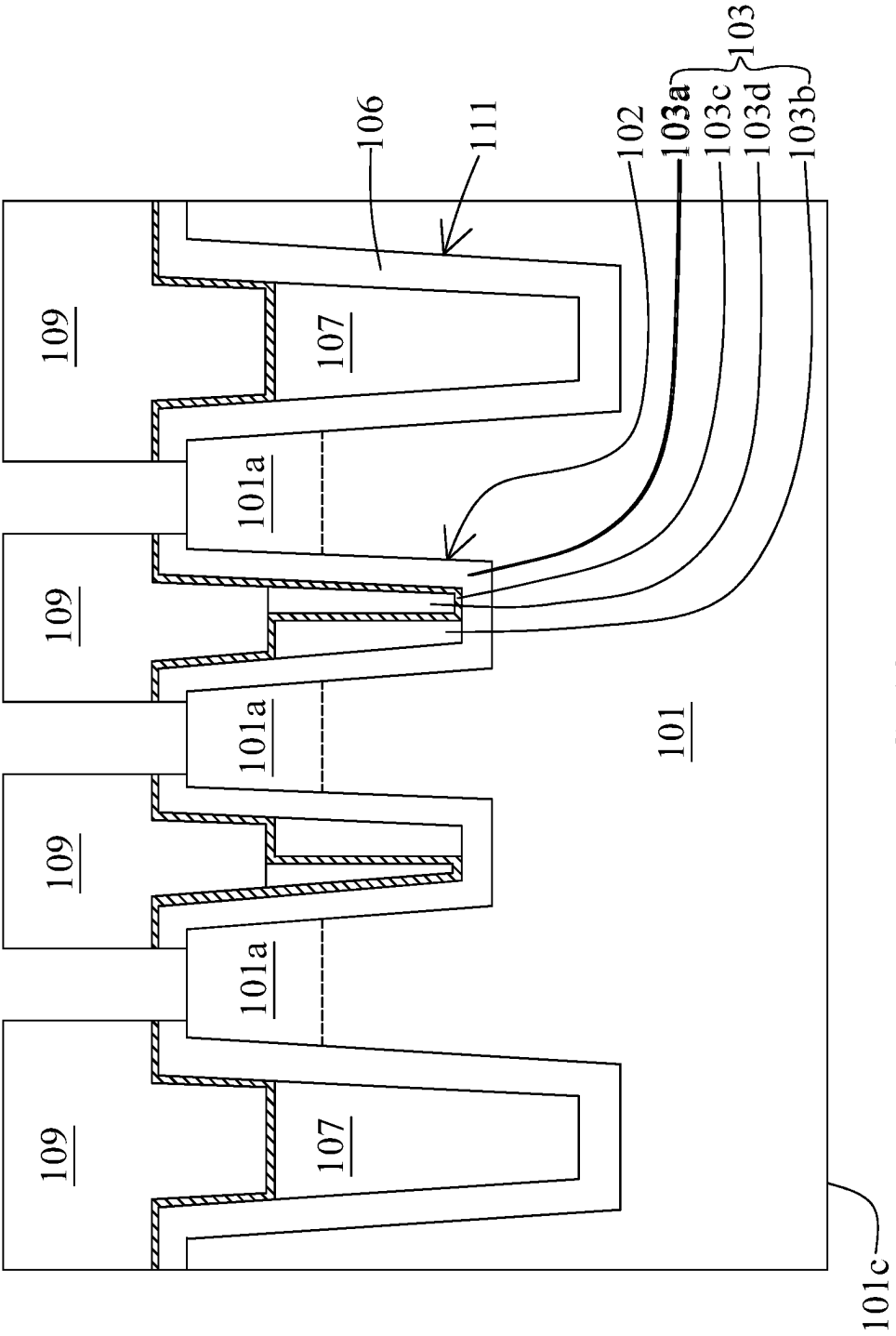


FIG. 18

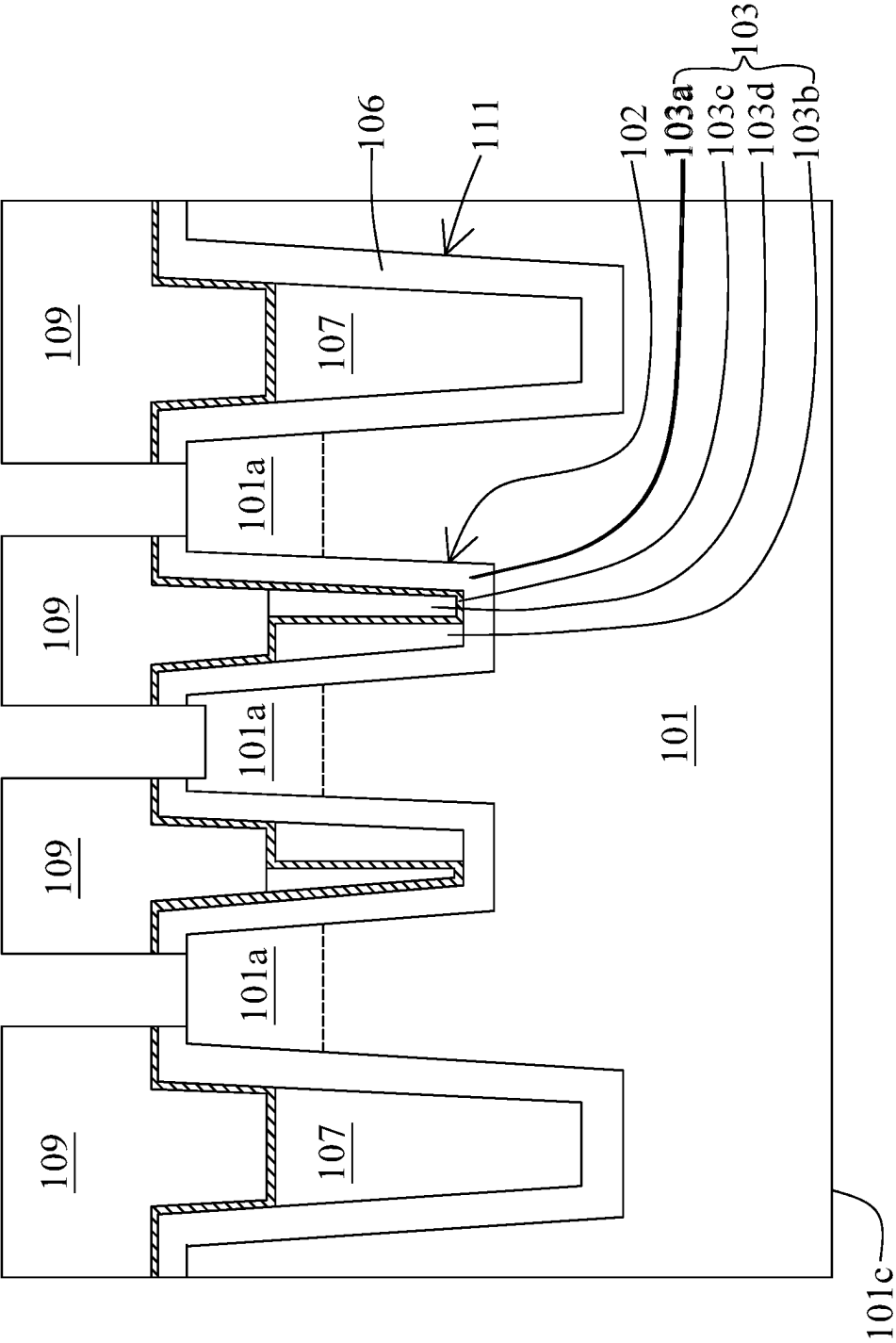


FIG. 19

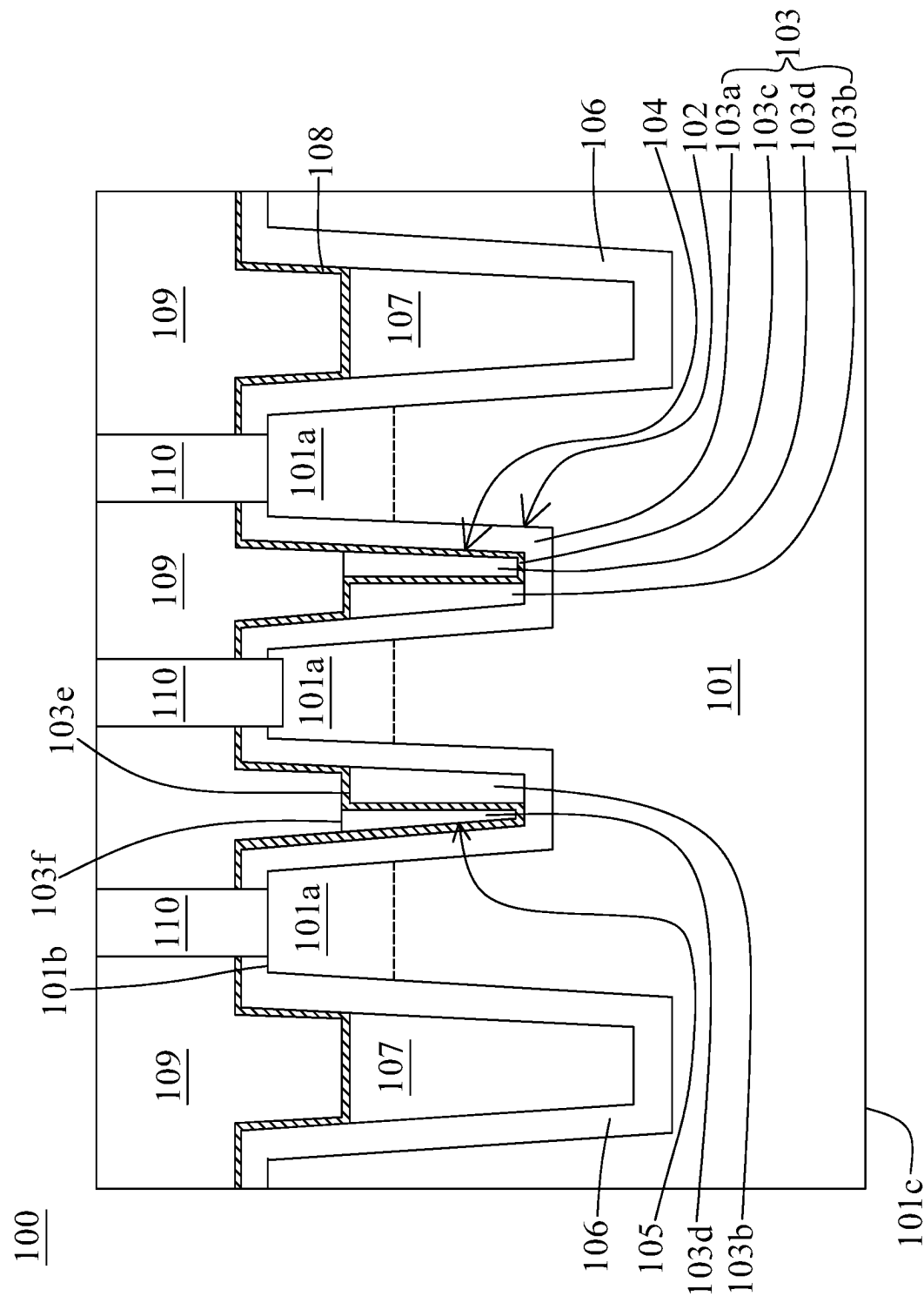


FIG. 20

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METHOD FOR MANUFACTURING MEMORY DEVICE HAVING WORD LINE WITH DUAL CONDUCTIVE MATERIALS

TECHNICAL FIELD

The present disclosure relates to a method for manufacturing a memory device, and more particularly, to a method for manufacturing a memory device having a word line (WL) with dual conductive materials.

DISCUSSION OF THE BACKGROUND

Dynamic random-access memory (DRAM) is a type of semiconductor arrangement for storing bits of data in separate capacitors within an integrated circuit (IC). DRAMs are commonly formed as trench capacitor DRAM cells. An advanced method of fabricating a buried gate electrode involves building a gate electrode of a transistor and a word line in a trench in an active area (AA) comprising a shallow trench isolation (STI) structure.

Over the past few decades, as semiconductor fabrication technology has continuously improved, sizes of electronic devices have been correspondingly reduced. As a size of a cell transistor is reduced to a few nanometers in length, current leakage may occur. The leakage may result in a significant drop in performance of the cell transistors. It is therefore desirable to develop improvements that address related manufacturing challenges.

SUMMARY

One aspect of the present disclosure provides a memory device. The memory device includes a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate, wherein the semiconductor substrate includes a recess extending from the surface into the semiconductor substrate; and a word line disposed within the recess, wherein the word line includes a first insulating layer disposed within and conformal to the recess, a first conductive member surrounded by the first insulating layer and disposed within the recess, a second insulating layer disposed conformal to the first insulating layer and the first conductive member, and a second conductive member disposed adjacent to the first conductive member and surrounded by the second insulating layer.

In some embodiments, a first work function of the first conductive member is substantially different from a second work function of the second conductive member.

In some embodiments, the first work function of the first conductive member is substantially greater than the second work function of the second conductive member.

In some embodiments, the first conductive member and the second conductive member include a same material.

In some embodiments, the first conductive member and the second conductive member include tungsten (W) or titanium nitride (TiN).

In some embodiments, the first insulating layer and the second insulating layer include oxide.

In some embodiments, the first conductive member is enclosed by the first insulating layer and the second insulating layer.

In some embodiments, the first conductive member is separated from the second conductive member by the second insulating layer.

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In some embodiments, a top surface of the first conductive member is substantially lower than a top surface of the second conductive member.

In some embodiments, the top surface of the first conductive member and the top surface of the second conductive member are surrounded by the active area of the semiconductor substrate.

In some embodiments, a width of the top surface of the first conductive member is substantially greater than or equal to a width of the top surface of the second conductive member.

In some embodiments, the memory device further comprises a dielectric layer disposed over the first conductive member, the second conductive member and the second insulating layer.

In some embodiments, the dielectric layer is in contact with a top surface of the second conductive member.

In some embodiments, the memory device further comprises a conductive plug extending through the dielectric layer and connecting to the active area of the semiconductor substrate.

In some embodiments, at least a portion of the second insulating layer is disposed between the dielectric layer and a top surface of the first conductive member.

In some embodiments, the dielectric layer includes nitride.

In some embodiments, a height of the first conductive member is substantially less than or equal to a height of the second conductive member.

Another aspect of the present disclosure provides a memory device. The memory device includes a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate, wherein the semiconductor substrate includes a first recess extending from the surface into the semiconductor substrate; and a word line disposed within the first recess, wherein the word line includes a first insulating layer disposed conformal to the first recess and having a second recess within the first recess, a first conductive member surrounded by the first insulating layer and disposed within the second recess, a second insulating layer disposed conformal to the second recess and the first conductive member and having a third recess within the second recess, and a second conductive member disposed within the third recess.

In some embodiments, the third recess has a first width at a position adjacent to the first conductive member and a second width at a position above the first conductive member and above the position having the first width.

In some embodiments, the second width is substantially different from the first width.

In some embodiments, the second width is substantially greater than the first width.

In some embodiments, a first work function of the first conductive member is substantially greater than a second work function of the second conductive member.

In some embodiments, the first work function of the first conductive member is substantially greater than 4 eV, and the second work function of the second conductive member is substantially less than 4 eV.

In some embodiments, a difference between the first work function and the second work function is substantially greater than 0.5 eV.

Another aspect of the present disclosure provides a method of manufacturing a memory device. The method includes steps of providing a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate; forming a recess extending from the sur-

face into the semiconductor substrate; disposing a first insulating layer conformal to the recess; disposing a first conductive material within the recess and surrounded by the first insulating layer; removing a portion of the first conductive material to form a first conductive member; disposing a second insulating layer within the recess and conformal to the first insulating layer and the first conductive member; and disposing a second conductive material within the recess and surrounded by the second insulating layer to form a second conductive member adjacent to the first conductive member.

In some embodiments, a first work function of the first conductive material is substantially different from a second work function of the second conductive material.

In some embodiments, the first work function of the first conductive material is substantially greater than the second work function of the second conductive member.

In some embodiments, the first conductive material is same as the second conductive material.

In some embodiments, the method further comprises disposing a patterned photoresist over the first insulating layer and the first conductive material, wherein the portion of the first conductive material exposed through the patterned photoresist is removed.

In some embodiments, the method further comprises removing the patterned photoresist after the formation of the first conductive member.

In some embodiments, a portion of the second conductive material disposed above the first conductive member is removed to form the second conductive member.

In some embodiments, the method further comprises disposing a dielectric layer over the first conductive member, the second conductive member and the second insulating layer; and forming a conductive plug extending through the dielectric layer and connecting to the active area of the semiconductor substrate.

In some embodiments, the disposing of the second insulating layer is performed prior to the disposing of the second conductive material.

In some embodiments, the disposing of the first conductive material is performed prior to the disposing of the second conductive material.

In some embodiments, the disposing of the first conductive material and the disposing of the second conductive material are performed separately.

In conclusion, because a word line includes dual conductive materials having different work functions, such difference can suppress or prevent a gate-induced drain leakage (GIDL). More specifically, because the word line includes a first conductive material with a high work function and a second conductive material with a low work function, such configuration can reduce the GIDL issue. Therefore, performance of the memory device and a process of manufacturing the memory device are improved.

The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a cross-sectional side view of a memory device in accordance with some embodiments of the present disclosure.

FIG. 2 is a flow diagram illustrating a method of manufacturing a memory device in accordance with some embodiments of the present disclosure.

FIGS. 3 to 20 illustrate cross-sectional views of intermediate stages in the formation of a memory device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

FIG. 1 is a schematic cross-sectional side view of a memory device 100 in accordance with some embodiments of the present disclosure. In some embodiments, the memory device 100 includes several unit cells arranged in rows and columns.

In some embodiments, the memory device 100 includes a semiconductor substrate 101. In some embodiments, the semiconductor substrate 101 includes semiconductive material such as silicon, germanium, gallium, arsenic, or a combination thereof. In some embodiments, the semiconductor substrate 101 includes bulk semiconductor material. In some embodiments, the semiconductor substrate 101 is a semiconductor wafer (e.g., a silicon wafer) or a semiconductor-on-insulator (SOI) wafer (e.g., a silicon-on-insulator wafer). In some embodiments, the semiconductor substrate 101 is a silicon substrate. In some embodiments, the semiconductor substrate 101 includes lightly-doped monocryst-

talline silicon. In some embodiments, the semiconductor substrate **101** is a p-type substrate.

In some embodiments, the semiconductor substrate **101** includes several active areas (AA) **101a**. The active area **101a** is a doped region in the semiconductor substrate **101**. In some embodiments, the active area **101a** extends horizontally over or under a top surface of the semiconductor substrate **101**. In some embodiments, the active area **101a** is disposed adjacent to the top surface of the semiconductor substrate **101**. In some embodiments, each of the active areas **101a** includes a same type of dopant. In some embodiments, each of the active areas **101a** includes a type of dopant that is different from types of dopants included in other active areas **101a**. In some embodiments, each of the active areas **101a** has a same conductive type. In some embodiments, the active area **101a** includes n-type dopants.

In some embodiments, the semiconductor substrate **101** includes a first surface **101b** and a second surface **101c** opposite to the first surface **101b**. In some embodiments, the first surface **101b** is a front side of the semiconductor substrate **101**, wherein electrical devices or components are subsequently formed over the first surface **101b** and configured to electrically connect to an external circuitry. In some embodiments, the active area **101a** is adjacent to or under the first surface **101b**. In some embodiments, the second surface **101c** is a back side of the semiconductor substrate **101**, where electrical devices or components are absent.

In some embodiments, the semiconductor substrate **101** includes a recess **102** extending into the semiconductor substrate **101**. In some embodiments, the recess **102** extends from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**. In some embodiments, the recess **102** is tapered from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**. In some embodiments, a depth of the recess **102** is substantially greater than a depth of the active area **101a**.

In some embodiments, the memory device **100** includes a word line **103** disposed within the recess **102**. In some embodiments, the word line **103** includes a first insulating layer **103a**, a first conductive member **103b**, a second insulating layer **103c** and a second conductive member **103d**.

In some embodiments, the first insulating layer **103a** is disposed within and conformal to the recess **102**. In some embodiments, at least a portion of the first insulating layer **103a** is surrounded by the active area **101a**. In some embodiments, the first insulating layer **103a** covers an entire sidewall of the recess **102**. In some embodiments, a portion of the first insulating layer **103a** is disposed over the first surface **101b** of the semiconductor substrate **101**. In some embodiments, the first insulating layer **103a** has a second recess **104** disposed within the recess **102**.

In some embodiments, the first insulating layer **103a** includes dielectric material such as oxide. In some embodiments, the first insulating layer **103a** is formed of an insulating material, such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof. In some embodiments, the first insulating layer **103a** includes dielectric material with a low dielectric constant (low k).

In some embodiments, the first conductive member **103b** is surrounded by the first insulating layer **103a** and disposed within the recess **102**. In some embodiments, the first conductive member **103b** extends within the first insulating layer **103a** from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**. In some embodiments, the first conductive member **103b** is disposed along a sidewall of the first insulating layer **103a**. In some

embodiments, the first conductive member **103b** is disposed within the second recess **104**.

In some embodiments, the first conductive member **103b** includes a first conductive material with a first work function. In some embodiments, the first work function of the first conductive member **103b** is substantially greater than 4 eV. In some embodiments, the first work function of the first conductive member **103b** is in a range of about 1 eV to about 8 eV. In some embodiments, the first conductive member **103b** includes conductive material such as titanium nitride (TiN), tungsten (W) or the like.

In some embodiments, the second insulating layer **103c** is disposed conformal to the first insulating layer **103a** and the first conductive member **103b**. In some embodiments, the second insulating layer **103c** is disposed conformal to a portion of the second recess **104**. In some embodiments, at least a portion of the second insulating layer **103c** is surrounded by the active area **101a**. In some embodiments, the second insulating layer **103c** and the first insulating layer **103a** enclose the first conductive member **103b**. In some embodiments, the second insulating layer **103c** covers the first conductive member **103b**.

In some embodiments, the second insulating layer **103c** is disposed along a portion of a sidewall of the first insulating layer **103a**, a portion of a bottom surface of the first insulating layer **103a**, a portion of a sidewall of the first conductive member **103b** and a top surface **103e** of the first conductive member **103b**. In some embodiments, a portion of the second insulating layer **103c** is disposed over the first surface **101b** of the semiconductor substrate **101**.

In some embodiments, the second insulating layer **103c** has a third recess **105** disposed within the second recess **104**. In some embodiments, the third recess **105** has a first width **W3** at a position adjacent to the first conductive member **103b** and a second width **W4** at a position above the first conductive member **103b** and above the position having the first width **W3**. In some embodiments, the second width **W4** is substantially different from the first width **W3**. In some embodiments, the second width **W4** is substantially greater than the first width **W3**.

In some embodiments, the second insulating layer **103c** includes a material different from a material of the first insulating layer **103a**. In some embodiments, the second insulating layer **103c** includes a same material as the first insulating layer **103a**. In some embodiments, the second insulating layer **103c** includes dielectric material such as oxide. In some embodiments, the second insulating layer **103c** is formed of an insulating material, such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof. In some embodiments, the second insulating layer **103c** includes dielectric material with a low dielectric constant (low k).

In some embodiments, the second conductive member **103d** is disposed adjacent to the first conductive member **103b** and surrounded by the second insulating layer **103c**. In some embodiments, the second conductive member **103d** extends within the second insulating layer **103c** from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**.

In some embodiments, the second conductive member **103d** extends substantially parallel to the first conductive member **103b**. The second conductive member **103d** is separated from the first conductive member **103b** by the second insulating layer **103c**. In some embodiments, the second conductive member **103d** is disposed within the third

recess **105**. In some embodiments, the first conductive member **103b** and the second conductive member **103d** serve as a gate electrode.

In some embodiments, the top surface **103e** of the first conductive member **103b** is substantially lower than a top surface **103f** of the second conductive member **103d**. In some embodiments, the top surface **103e** of the first conductive member **103b** and the top surface **103f** of the second conductive member **103d** are surrounded by the active area **101a** of the semiconductor substrate **101**. In some embodiments, the top surface **103f** of the second conductive member **103d** is substantially coplanar with a horizontal surface of the second insulating layer **103c**.

In some embodiments, a width **W1** of the top surface **103e** of the first conductive member **103b** is substantially greater than or equal to a width **W2** of the top surface **103f** of the second conductive member **103d**. In some embodiments, a height **H1** of the first conductive member **103b** is substantially less than or equal to a height **H2** of the second conductive member **103d**.

In some embodiments, the second conductive member **103d** includes a second conductive material with a second work function. In some embodiments, the first work function of the first conductive member **103b** is substantially different from the second work function of the second conductive member **103d**. In some embodiments, the first work function of the first conductive member **103b** is substantially greater than the second work function of the second conductive member **103d**.

In some embodiments, the second work function of the second conductive member **103d** is substantially less than 4 eV. In some embodiments, the second work function of the second conductive member **103d** is in a range of about 0.1 eV to about 1 eV. In some embodiments, a difference between the first work function of the first conductive member **103b** and the second work function of the second conductive member **103d** is substantially greater than 0.5 eV. In some embodiments, the second conductive member **103d** includes conductive material such as titanium nitride (TiN), tungsten (W) or the like. In some embodiments, the second conductive member **103d** includes a material same as a material of the first conductive member **103b**.

In some embodiments, the memory device **100** further includes an isolation structure **106** adjacent to the word line **103**. In some embodiments, the isolation structure **106** extends into the semiconductor substrate **101** from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**.

In some embodiments, the isolation structure **106** is a shallow trench isolation (STI). In some embodiments, the isolation structure **106** defines a boundary of the active area **101a**. In some embodiments, the isolation structure **106** includes dielectric material such as oxide. In some embodiments, the isolation structure **106** is formed of an insulating material, such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof.

In some embodiments, a third conductive member **107** is surrounded by the isolation structure **106**. In some embodiments, the third conductive member **107** has a third work function substantially equal to the first work function of the first conductive member **103b**. In some embodiments, the third work function of the third conductive member **107** is substantially greater than 4 eV. In some embodiments, the third conductive member **107** includes conductive material such as titanium nitride (TiN), tungsten (W) or the like.

In some embodiments, a third insulating layer **108** is disposed over the third conductive member **107** and the

isolation structure **106**. In some embodiments, the third insulating layer **108** includes a material same as the material of the second insulating layer **103c**. In some embodiments, the third insulating layer **108** includes dielectric material such as oxide. In some embodiments, the third insulating layer **108** is formed of an insulating material, such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof.

In some embodiments, the memory device **100** further includes a dielectric layer **109** disposed over the first conductive member **103b**, the second conductive member **103d**, the second insulating layer **103c**, the third conductive member **107** and the third insulating layer **108**. In some embodiments, the dielectric layer **109** is in contact with the second insulating layer **103c**, the third insulating layer **108** and the top surface **103f** of the second conductive member **103d**.

In some embodiments, at least a portion of the second insulating layer **103c** is disposed between the dielectric layer **109** and the top surface **103e** of the first conductive member **103b**. In some embodiments, the dielectric layer **109** includes dielectric material such as nitride. In some embodiments, the dielectric layer **109** serves as a gate dielectric.

In some embodiments, the memory device **100** further includes a conductive plug **110** extending through the dielectric layer **109** and connecting to the active area **101a** of the semiconductor substrate **101**. In some embodiments, the conductive plug **110** extends through the first insulating layer **103a** and the second insulating layer **103c**. In some embodiments, a portion of the conductive plug **110** protrudes into the semiconductor substrate **101** or the active area **101a** of the semiconductor substrate **101**, such that the portion of the conductive plug **110** is surrounded by the semiconductor substrate **101** or the active area **101a** of the semiconductor substrate **101**. In some embodiments, the conductive plug **110** includes conductive material such as metal. In some embodiments, the conductive plug **110** includes copper, gold, silver or the like.

Since the word line **103** includes dual conductive materials having different work functions, such difference can suppress or prevent a gate-induced drain leakage (GIDL). More specifically, because the word line **103** includes the first conductive member **103b** with the high work function and the second conductive member **103d** with the low work function, such configuration can reduce the GIDL issue. Therefore, performance of the memory device **100** can be improved.

FIG. 2 is a flow diagram illustrating a method **S200** of manufacturing a memory device **100** in accordance with some embodiments of the present disclosure, and FIGS. 3 to 20 illustrate cross-sectional views of intermediate stages in formation of the memory device **100** in accordance with some embodiments of the present disclosure.

The stages shown in FIGS. 3 to 20 are also illustrated schematically in the flow diagram in FIG. 2. In following discussion, the fabrication stages shown in FIGS. 3 to 20 are discussed in reference to process steps shown in FIG. 2. The method **S200** includes a number of operations, and description and illustration are not deemed as a limitation to a sequence of the operations. The method **S200** includes a number of steps (**S201**, **S202**, **S203**, **S204**, **S205**, **S206** and **S207**).

Referring to FIGS. 3 to 5, a semiconductor substrate **101** with an active area **101a** defined adjacent to a first surface **101b** of the semiconductor substrate **101** is provided according to step **S201** in FIG. 2. In some embodiments, the semiconductor substrate **101** having the first surface **101b** and a second surface **101c** opposite to the first surface **101b**

is provided as shown in FIG. 3. In some embodiments, a trench **111** extending from the first surface **101b** toward the second surface **101c** is formed as shown in FIG. 4. The trench **111** is formed by removing some portions of the semiconductor substrate **101**.

In some embodiments, the semiconductor substrate **101** includes an isolation structure **106** surrounding the active area **101a**. In some embodiments, the isolation structure **106** extends from the first surface **101b** toward the second surface **101c** of the semiconductor substrate **101**. In some embodiments, the isolation structure **106** includes dielectric material such as oxide or the like. In some embodiments, the isolation structure **106** is formed by disposing an isolation material into the trench **111** as shown in FIG. 5.

Referring to FIG. 6, a recess **102** extending from the first surface **101b** into the semiconductor substrate **101** is formed according to step **S202** in FIG. 2. In some embodiments, the recess **102** is formed by removing some portions of the semiconductor substrate **101**. In some embodiments, the portions of the semiconductor substrate **101** are removed by etching or any other suitable process. In some embodiments, the recess **102** is at least partially surrounded by the active area **101a** of the semiconductor substrate **101**. In some embodiments, a portion of the isolation structure **106** is removed.

Referring to FIG. 7, a first insulating layer **103a** conformal to the recess **102** is disposed according to step **S203** in FIG. 2. In some embodiments, the first insulating layer **103a** is disposed over the first surface **101b** of the semiconductor substrate **101**. In some embodiments, the first insulating layer **103a** covers an entire sidewall of the recess **102**. In some embodiments, the first insulating layer **103a** includes dielectric material such as oxide. In some embodiments, the first insulating layer **103a** and the isolation structure **106** include a same material. In some embodiments, the first insulating layer **103a** is formed by deposition, oxidation or any other suitable process.

Referring to FIG. 8, a first conductive material **112** is disposed according to step **S204** in FIG. 2. In some embodiments, the first conductive material **112** is disposed within the recess **102** and surrounded by the first insulating layer **103a**. In some embodiments, the first conductive material **112** is also disposed within the trench **111** and surrounded by the isolation structure **106**. In some embodiments, at least a portion of the first conductive material **112** is surrounded by the active area **101a** of the semiconductor substrate **101**. In some embodiments, the first conductive material **112** is disposed by deposition, chemical vapor deposition (CVD) or any other suitable process.

In some embodiments, the first conductive material **112** has a first work function substantially greater than 4 eV. In some embodiments, the first conductive material **112** is titanium nitride (TiN), tungsten (W) or the like.

Referring to FIGS. 9 to 12, a portion of the first conductive material **112** is removed to form a first conductive member **103b** according to step **S205** in FIG. 2. In some embodiments, a photoresist **114'** is disposed over the first conductive material **112**, the first insulating layer **103a** and the isolation structure **106** as shown in FIG. 9. In some embodiments, the photoresist **114'** is disposed by spin coating or any other suitable process.

In some embodiments, after the disposing of the photoresist **114'**, a portion of the photoresist **114'** is removed to form a patterned photoresist **114** as shown in FIG. 10. In some embodiments, a portion of the first conductive material **112** is exposed through the patterned photoresist **114**. In some embodiments, the portion of the first conductive

material **112** is disposed within the recess **102** and surrounded by the first insulating layer **103a**.

In some embodiments, the portion of the first conductive material **112** exposed through the patterned photoresist **114** is removed as shown in FIG. 11. In some embodiments, the portion of the first conductive material **112** exposed through the patterned photoresist **114** is removed by etching or any other suitable process.

In some embodiments, after the removal of the portion of the first conductive material **112** exposed through the patterned photoresist **114**, the patterned photoresist **114** is removed and the first conductive member **103b** is formed as shown in FIG. 12. In some embodiments, a top surface **103e** of the first conductive member **103b** is formed. In some embodiments, the patterned photoresist **114** is removed by etching, stripping or any other suitable process.

In some embodiments, a third conductive member **107** is also formed within the trench **111** and surrounded by the isolation structure **106**. In some embodiments, the first conductive member **103b** and the third conductive member **107** are formed simultaneously or separately.

Referring to FIG. 13, a second insulating layer **103c** is disposed within the recess **102** and conformal to the first insulating layer **103a** and the first conductive member **103b** according to step **S206** in FIG. 2. In some embodiments, the second insulating layer **103c** covers the first conductive member **103b** and the third conductive member **107**. In some embodiments, the second insulating layer **103c** is also disposed over the first surface **101b** of the semiconductor substrate **101**.

In some embodiments, the first conductive member **103b** is enclosed by the second insulating layer **103c** and the first insulating layer **103a**. In some embodiments, the second insulating layer **103c** includes dielectric material such as oxide. In some embodiments, the second insulating layer **103c** is formed by deposition, atomic layer deposition (ALD) or any other suitable process.

Referring to FIGS. 14 and 15, a second conductive material **113** is disposed within the recess **102** and surrounded by the second insulating layer **103c** to form a second conductive member **103d** adjacent to the first conductive member **103b** according to step **S207** in FIG. 2. In some embodiments, the second conductive material **113** is disposed over the second insulating layer **103c** as shown in FIG. 14. In some embodiments, the second conductive material **113** is disposed by deposition, chemical vapor deposition (CVD) or any other suitable process.

In some embodiments, the disposing of the second insulating layer **103c** is performed prior to the disposing of the second conductive material **113**. In some embodiments, the disposing of the first conductive material **112** is performed prior to the disposing of the second conductive material **113**. In some embodiments, the disposing of the first conductive material **112** and the disposing of the second conductive material **113** are performed separately.

In some embodiments, some portions of the second conductive material **113** are removed to form the second conductive member **103d** as shown in FIG. 15. In some embodiments, some portions of the second conductive material **113** are removed by etching or any other suitable process. In some embodiments, a portion of the second conductive material **113** disposed above the first conductive member **103b** is removed to form the second conductive member **103d**.

In some embodiments, a portion of the second conductive material **113** above the third conductive member **107** is removed. In some embodiments, a portion of the second

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conductive material **113** is removed until a top surface **103f** of the second conductive member **103d** is substantially coplanar with a surface of a horizontal portion of the second insulating layer **103c**.

In some embodiments, the top surface **103e** of the first conductive member **103b** is substantially lower than the top surface **103f** of the second conductive member **103d**. In some embodiments, the top surface **103e** of the first conductive member **103b** and the top surface **103f** of the second conductive member **103d** are surrounded by the active area **101a** of the semiconductor substrate **101**.

In some embodiments, the second conductive material **113** has a second work function substantially different from the first work function of the first conductive material **112**. In some embodiments, the first work function of the first conductive material **112** is substantially greater than the second work function of the second conductive material **113**. In some embodiments, the second work function of the second conductive material **113** is substantially less than 4 eV.

In some embodiments, a difference between the first work function of the first conductive material **112** and the second work function of the second conductive material **113** is substantially greater than 0.5 eV. In some embodiments, the second conductive material **113** is titanium nitride (TiN), tungsten (W) or the like. In some embodiments, the second conductive material **113** includes a material same as a material of the first conductive material **112**.

In some embodiments, after the formation of the second conductive member **103d**, a dielectric material **109'** is disposed over the second insulating layer **103c** and the second conductive member **103d** as shown in FIG. 16. In some embodiments, the dielectric material **109'** is disposed by deposition, CVD or any other suitable process. In some embodiments, the dielectric material **109'** is nitride or the like.

In some embodiments, after the disposing of the dielectric material **109'**, a portion of the dielectric material **109'** is removed to form a dielectric layer **109** as shown in FIG. 17. In some embodiments, the portion of the dielectric material **109'** is removed by etching or any other suitable process. In some embodiments, a portion of the second insulating layer **103c** is exposed through the dielectric layer **109**.

In some embodiments, the portion of the second insulating layer **103c** exposed through the dielectric layer **109** is removed as shown in FIG. 18. In some embodiments, a portion of the first insulating layer **103a** exposed through the dielectric layer **109** is also removed. In some embodiments, the removal of the portion of the second insulating layer **103c** and the portion of the first insulating layer **103a** exposed through the dielectric layer **109** is implemented by etching or any other suitable process. In some embodiments, a portion of the semiconductor substrate **101** exposed through the dielectric layer **109** is further removed as shown in FIG. 19.

In some embodiments, after the removal of the portion of the first insulating layer **103a** and the portion of the second insulating layer **103c** exposed through the dielectric layer **109**, a conductive plug **110** is formed over the active area **101a** of the semiconductor substrate **101** as shown in FIG. 20. In some embodiments, the conductive plug **110** is formed by disposing a conductive material. In some embodiments, the conductive material is disposed by electroplating or any other suitable process.

In some embodiments, the conductive plug **110** extends through the dielectric layer **109** and connects to the active area **101a** of the semiconductor substrate **101**. In some

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embodiments, the conductive plug **110** is electrically connected to the word line **103** through the active area **101a** of the semiconductor substrate **101**. In some embodiments, the memory device **100** of FIG. 1 is formed as shown in FIG. 20.

In an aspect of the present disclosure, a memory device is provided. The memory device includes a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate, wherein the semiconductor substrate includes a recess extending from the surface into the semiconductor substrate; and a word line disposed within the recess,

wherein the word line includes a first insulating layer disposed within and conformal to the recess, a first conductive member surrounded by the first insulating layer and disposed within the recess, a second insulating layer disposed conformal to the first insulating layer and the first conductive member, and a second conductive member disposed adjacent to the first conductive member and surrounded by the second insulating layer.

In another aspect of the present disclosure, a memory device is provided. The memory device includes a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate, wherein the semiconductor substrate includes a first recess extending from the surface into the semiconductor substrate; and a word line disposed within the first recess, wherein the word line includes a first insulating layer disposed conformal to the first recess and having a second recess within the first recess, a first conductive member surrounded by the first insulating layer and disposed within the second recess, a second insulating layer disposed conformal to the second recess and the first conductive member and having a third recess within the second recess, and a second conductive member disposed within the third recess.

In another aspect of the present disclosure, a method of manufacturing a memory device is provided. The method includes steps of providing a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate; forming a recess extending from the surface into the semiconductor substrate; disposing a first insulating layer conformal to the recess; disposing a first conductive material within the recess and surrounded by the first insulating layer; removing a portion of the first conductive material to form a first conductive member; disposing a second insulating layer within the recess and conformal to the first insulating layer and the first conductive member; and disposing a second conductive material within the recess and surrounded by the second insulating layer to form a second conductive member adjacent to the first conductive member.

In conclusion, because a word line includes dual conductive materials having different work functions, such difference can suppress or prevent a gate-induced drain leakage (GIDL). More specifically, because the word line includes a first conductive material with a high work function and a second conductive material with a low work function, such configuration can reduce the GIDL issue. Therefore, performance of the memory device and process of manufacturing of the memory device are improved.

Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

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Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein, may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods and steps.

What is claimed is:

1. A method of manufacturing a memory device, comprising:

providing a semiconductor substrate with an active area defined adjacent to a surface of the semiconductor substrate;

forming a first recess extending from the surface into the semiconductor substrate;

disposing a first insulating layer conformal to the first recess, wherein the first insulating layer covers an entire sidewall of the first recess and forms a second recess surrounded by the first insulating layer;

disposing a first conductive material within the second recess and surrounded by the first insulating layer;

removing a portion of the first conductive material to form a first conductive member and to expose an upper surface of the first insulating layer in proximity to a bottom of the second recess;

disposing a second insulating layer within the second recess and conformal to the first insulating layer and the first conductive member, wherein the second insulating layer is coupled to the upper surface of the first insulating layer in proximity to the bottom of the second recess; and

disposing a second conductive material within the second recess and surrounded by the second insulating layer to form a second conductive member adjacent to the first conductive member,

wherein the second conductive member is separated from the first insulating layer and the first conductive member by the second insulating layer.

2. The method according to claim 1, wherein a first work function of the first conductive material is substantially different from a second work function of the second conductive material.

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3. The method according to claim 2, wherein the first work function of the first conductive material is substantially greater than the second work function of the second conductive material.

4. The method according to claim 1, wherein the first conductive material is the same as the second conductive material.

5. The method according to claim 1, further comprising disposing a patterned photoresist over the first insulating layer and the first conductive material, wherein the portion of the first conductive material exposed through the patterned photoresist is removed.

6. The method according to claim 5, further comprising removing the patterned photoresist after the formation of the first conductive member.

7. The method according to claim 1, wherein a portion of the second conductive material disposed above the first conductive member is removed to form the second conductive member.

8. The method according to claim 1, further comprising: disposing a dielectric layer over the first conductive member, the second conductive member and the second insulating layer; and

forming a conductive plug extending through the dielectric layer and connecting to the active area of the semiconductor substrate.

9. The method according to claim 1, wherein the disposing of the second insulating layer is performed prior to the disposing of the second conductive material.

10. The method according to claim 1, wherein the disposing of the first conductive material is performed prior to the disposing of the second conductive material.

11. The method according to claim 1, wherein the disposing of the first conductive material and the disposing of the second conductive material are performed separately.

12. The method according to claim 2, wherein the first work function of the first conductive material is substantially greater than 4 eV, and the second work function of the second conductive material is substantially less than 4 eV.

13. The method according to claim 2, wherein a difference between the first work function and the second work function is substantially greater than 0.5 eV.

14. The method according to claim 1, wherein a top surface of the first conductive member is substantially lower than a top surface of the second conductive member.

15. The method according to claim 14, wherein the top surface of the first conductive member and the top surface of the second conductive member are surrounded by the active area of the semiconductor substrate.

16. The method according to claim 14, wherein a width of the top surface of the first conductive member is substantially greater than or equal to a width of the top surface of the second conductive member.

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